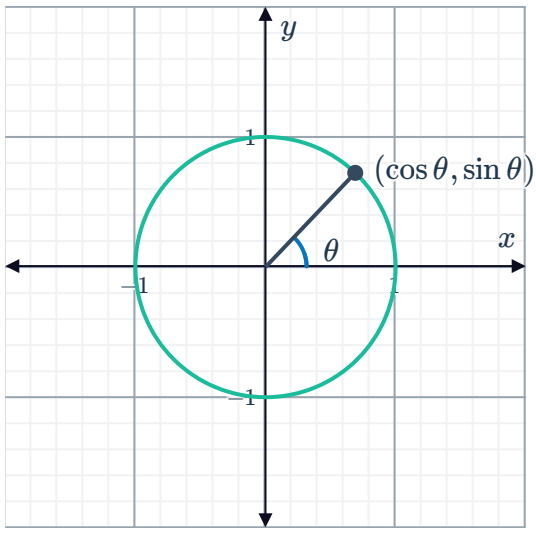
# Graphs of trigonometric functions

**1** The value of $\sin\theta$ can be represented on the given $xy$-plane.

**a** Consider the curve $y = \sin\theta$:



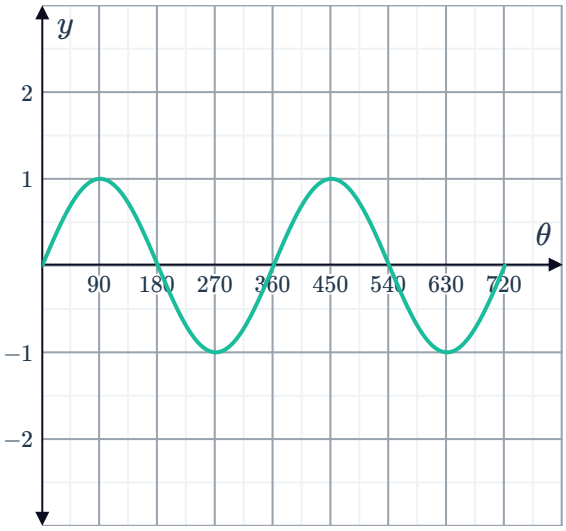

Complete the following:

**i** A rotation of $380^\circ$ has the same rotation as the acute angle with measure ⬚. So $\sin 380^\circ = \sin$ ⬚.

**ii** A rotation of $480^\circ$ has the same rotation as the obtuse angle with measure ⬚. So $\sin 480^\circ = \sin$ ⬚.

**iii** A rotation of $580^\circ$ has the same rotation as the reflex angle with measure ⬚. So $\sin 580^\circ = \sin$ ⬚$^\circ$.

**b** Consider the graph of $y = \sin\theta$ for $0 \leq \theta \leq 720^\circ$:

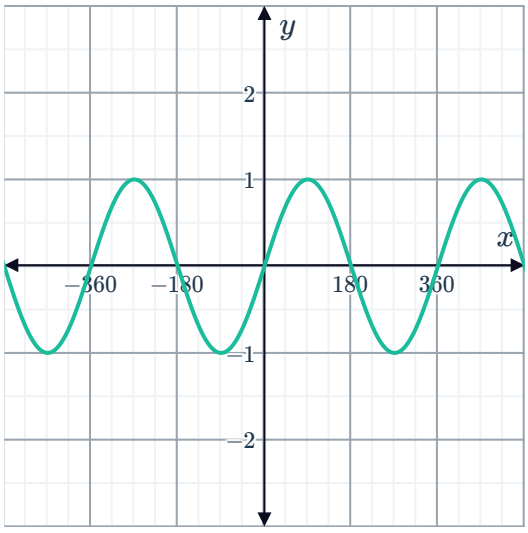



Describe the nature of $y = \sin\theta$.

**c** The number of degrees it takes for the curve to complete a full cycle is called the period of the function.

Determine the period of $y = \sin\theta$.

**2** Consider the curve $y = \sin x$ and state whether the following statements are true or false:




**a** The graph of $y = \sin x$ is periodic.

**b** As $x$ approaches infinity, the height of the graph approaches infinity.

**c** The graph of $y = \sin x$ is increasing between $x = (-90)^\circ$ and $x = 0^\circ$.

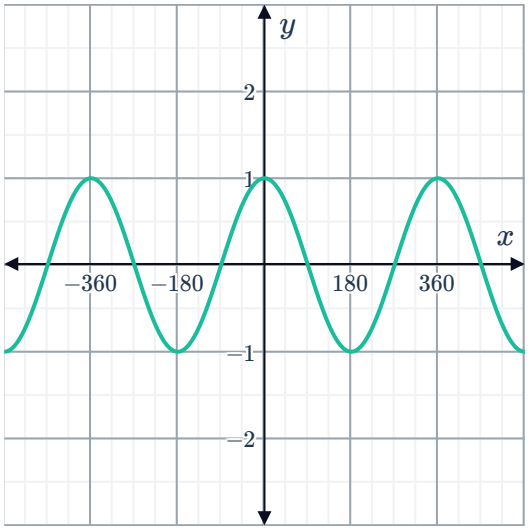
**d** The graph of $y = \sin x$ is symmetric about the line $x = 0$.

**e** The graph of $y = \sin x$ is symmetric with respect to the origin.

**f** The $y$-values of the graph repeat after a period of $360^\circ$.

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**3** Consider the curve $y = \cos x$ and state whether the following statements are true or false:




**a** The graph of $y = \cos x$ is cyclic.

**b** As $x$ approaches infinity, the height of the graph approaches infinity.

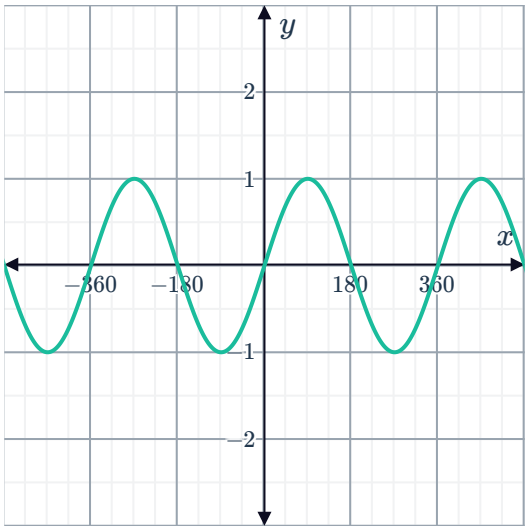
**c** The graph of $y = \cos x$ is increasing between $x = (-180)^\circ$ and $x = (-90)^\circ$.

**d** The graph of $y = \cos x$ has reflective symmetry across the line $x = 0$.

**e** The graph of $y = \cos x$has rotational symmetric with respect to the origin.

**f** The $y$-values of the graph repeat after a period of $180^\circ$.

**4** Consider the curve $y = \sin x$:

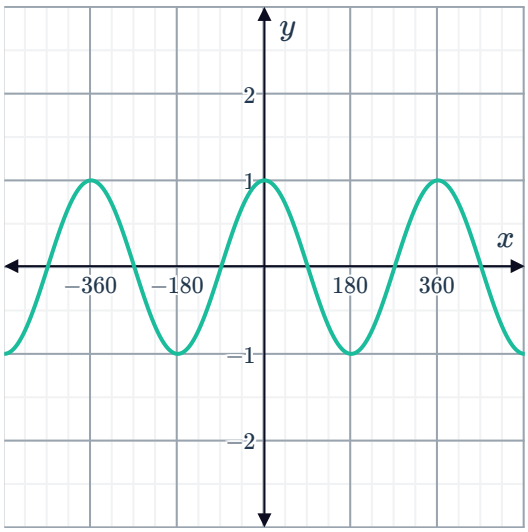



**a** State the coordinates of the $y$-intercept.

**b** Find the maximum $y$-value.

**c** Find the minimum $y$-value.

**5** Consider the curve $y = \cos x$:




**a** State the coordinates of the $y$-intercept.

**b** Find the maximum $y$-value.

**c** Find the minimum $y$-value.

**6** Consider the equation $y = \cos x$

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**a** Complete the table of values:

**b** Sketch the graph of $y = \cos x$.

| $x$ | $0^\circ$ | $90^\circ$ | $180^\circ$ | $270^\circ$ | $360^\circ$ |
|---|---|---|---|---|---|
| $\cos x$ | | | | | |

**7** Consider the equation $y = \sin x$.

**a** Complete the table of values:

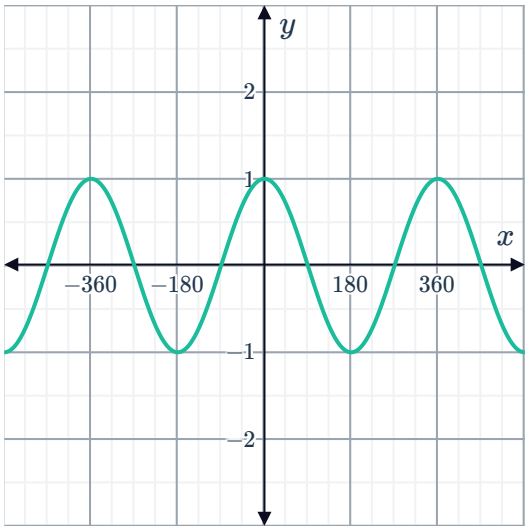
**b** Sketch the graph of $y = \sin x$.

| $x$ | $0^\circ$ | $90^\circ$ | $180^\circ$ | $270^\circ$ | $360^\circ$ |
|---|---|---|---|---|---|
| $\sin x$ | | | | | |

**8** Consider the graph of $y = \cos x$:

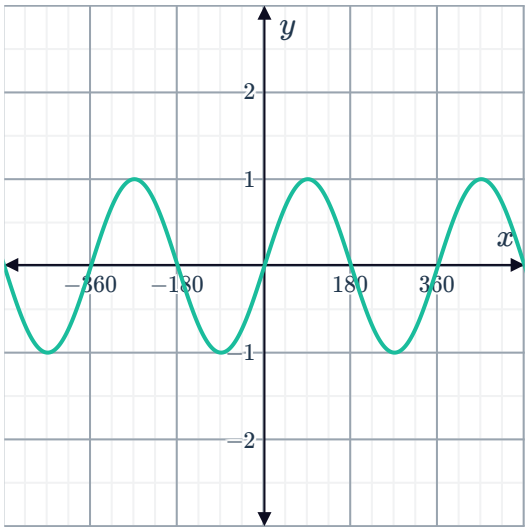
**a** Is $\cos 162^\circ$ positive or negative?

**b** In which quadrant does an angle with measure $162^\circ$ lie in?




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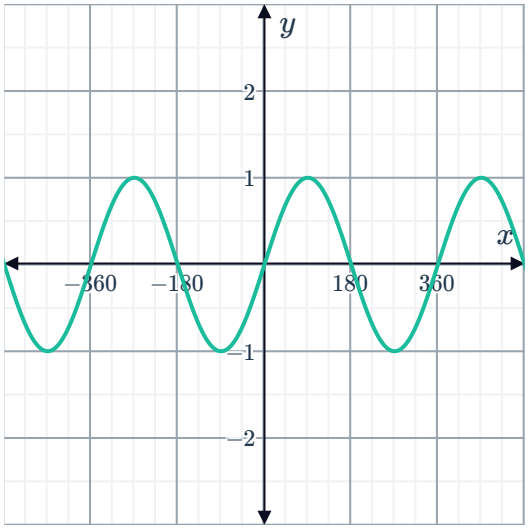
**9** Consider the graph of $y = \sin x$:




**a** Is $\sin 200^\circ$ positive or negative?

**b** In which quadrant does an angle with measure $200^\circ$ lie in?

**10** Consider the curve $y = \sin x$:




**a** If one cycle of the graph of $y = \sin x$ starts at $x = 0^\circ$, when does the next cycle start?

**b** Find the $x$-value of the $x$-intercept in the following regions:

**i** $0^\circ < x < 360^\circ$

**ii** $(-360)^\circ < x < 0^\circ$

**c** True or false: As $x$ approaches infinity, the graph of $y = \sin x$ stays between $y = -1$ and $y = 1$.

**d** State whether the graph of $y = \sin x$ increases or decreases in the following regions:

**i** $90^\circ < x < 270^\circ$

**ii** $(-270)^\circ < x < (-90)^\circ$

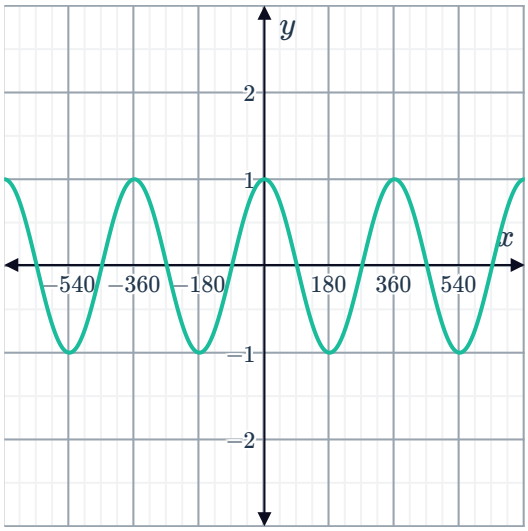
**iii** $(-450)^\circ < x < (-270)^\circ$

**iv** $(-90)^\circ < x < 90^\circ$

**v** $270^\circ < x < 450^\circ$

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**11** Consider the curve $y = \cos x$:




**a** If one cycle of the graph of $y = \cos x$ starts at $x = (-90)^\circ$, when does the next cycle start?

**b** Find the $x$-values of the $x$-intercepts in the following regions:

i $0^\circ < x < 360^\circ$

ii $(-360)^\circ < x < 0^\circ$

**c** True or false: As $x$ approaches infinity, the graph of $y = \cos x$ stays between $y = -1$ and $y = 1$.

**d** State whether the graph of $y = \cos x$ increases or decreases in the following regions:

i $0^\circ < x < 180^\circ$

ii $(-180)^\circ < x < 0^\circ$

iii $180^\circ < x < 360^\circ$

iv $(-360)^\circ < x < (-180)^\circ$

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